

# Predicting TCR Cross-Reactivity Before It Reaches a Patient

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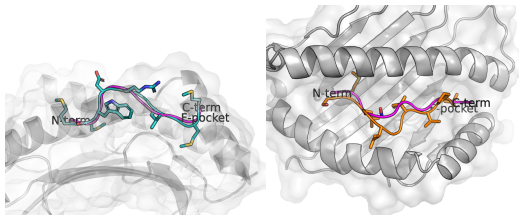


Figure 1: **Proposed framework.** A two-state discrimination test for structure-based peptide design in a cross-reactive pMHC–TCR system.

## 1 Motivation

TCR-engineering therapies promise a transformative future for cancer treatment, yet cross-reactivity remains a fundamental barrier to scaling them into personalized medicine. The danger is not hypothetical: an affinity-enhanced MAGE-A3 receptor that had cleared extensive preclinical testing caused fatal cardiac toxicity in two patients, cross-recognizing an unrelated self-peptide from the muscle protein titin that was identified only after the fact.

Three problems keep this liability beyond the reach of current methods. First, experimental validation is slow and costly, an untenable price when the presented peptide space is astronomically large (on the order of  $10^{11}$ , roughly the number of stars in the Milky Way) and the fast-growing tumors these therapies target leave only a narrow clinical window to act. Second, the extreme polymorphism of HLA leaves binding data sparse for most alleles and for underrepresented populations, precisely where personalized therapy is needed most. Third, amino-acid scanning resolves one receptor at a time and does not scale to the combinatorics of the peptide–TCR interaction.

## 2 Proposed Framework

I would bring two complementary tools of my own design to this problem: **ESMCBA**, a binding-affinity system built on protein language models with domain adaptation that recovers usable signal even for data-scarce MHC alleles (Mares, Espinoza, and Ioannidis 2025a); and a **structure-guided generative framework** that uses diffusion models conditioned on the MHC groove to sample compatible peptides across high-priority alleles (Mares, Espinoza, and Ioannidis 2025b). Together they let me computationally screen these interactions and chart the cross-reactive neighborhood of a receptor before it ever reaches a patient.

Making that screen trustworthy requires an evaluation standard the field lacks. Generative backbone models are scored by deviation to a single crystal structure, which cannot separate a design genuinely matching a target conformation from one merely close to the groove; sequence models are summarized by one average recovery rate, obscuring which constraints produce the signal. I propose that every

claim instead be evaluated by discrimination between two structurally defined alternatives (Figure 1).

**Aim 1: Backbone conformation.** Cross-reactive TCRs engage chemically distinct peptides through distinct bound conformations, giving a natural two-state system in which off-target recognition is structurally explicit. I will ask whether a generative model, given only receptor-side contacts and no peptide template, can be steered to a target conformation, including deliberately to the conformation *other* than the one nearest its conditioning input. Thresholds will be calibrated against native molecular-dynamics ensembles rather than a single snapshot, and independent criteria (proximity, anchor identity, burial depth) must agree.

**Aim 2: Sequence constraints.** Inverse-folding models recover native TCR interface residues well above chance (Ribeiro-Filho et al. 2024), but on the receptor side. I will apply them to the *peptide* across a panel of solved pMHC–TCR structures, with the TCR present and computationally removed, to identify which positions a receptor actually reads. Those positions define the axes along which a cross-reactive partner can differ, and therefore where a safety screen must look.

## 3 Expected Outcomes

This fellowship would let me develop that screen into a routine safeguard, turning cross-reactivity from a lethal surprise into a predictable and testable step, and making the next generation of cancer therapies measurably safer.

## References

- Mares, S.; Espinoza, A.; and Ioannidis, N. M. 2025a. Continued domain-specific pre-training of protein language models for pMHC-I binding prediction. *Proc. 20th Machine Learning in Computational Biology*, PMLR 311:304–325.
- Mares, S.; Espinoza, A.; and Ioannidis, N. M. 2025b. Generation of structure-guided pMHC-I libraries using diffusion models. *ICML Generative AI and Biology Workshop*.
- Ribeiro-Filho, H. V.; et al. 2024. Exploring the potential of structure-based deep learning approaches for T cell receptor design. *PLOS Comput. Biol.*